

RIGOR-OMR: Relation-Aware Interpretable Geometry for Zero-Shot Cross-Domain Optical Music Recognition

Anonymous submission

Abstract

Optical music recognition (OMR) systems achieve strong performance on in-domain benchmarks, yet whether such results translate into plug-and-play recognition on unseen score domains remains underexplored. We present RIGOR-OMR (Relation-aware Interpretable Geometry for Out-of-domain Recognition), a modular framework that leverages staff-normalized geometry to guide visual symbol detection and explicit relation reasoning, while reconstructing structured outputs without target-domain training or dataset-specific adaptation. To compare systems with heterogeneous outputs, we introduce a representation-agnostic pitch–duration event evaluation protocol. On Polish Scores, RIGOR-OMR achieves 17.083 Event-F1, representing relative improvements of 357.8% over SMT and 1182.4% over Zeus. On Debussy, RIGOR-OMR significantly outperforms SMT while performing comparably to Zeus. Ablation studies confirm that explicit relation reasoning is the primary source of these improvements. Extensive experiments across Polish Scores, Debussy, GrandStaff, and OLiMPiC establish RIGOR-OMR as an effective approach to zero-shot music-content recognition under substantial domain shift.

1 Introduction

Optical music recognition converts score images into machine-readable notation. Unlike ordinary character recognition, the meaning of a mark depends jointly on its class, staff-relative position, attachment relations, voice, and reading order (Calvo-Zaragoza, Jr., and Pacha 2021; Rebelo et al. 2012). Modern systems increasingly treat OMR as image-to-sequence prediction, using Transformer decoders to emit compact musical encodings (Vaswani et al. 2017; Ríos-Vila, Calvo-Zaragoza, and Paquet 2024; Li et al. 2023). This formulation has enabled recognition beyond monophonic incipits and toward piano systems or complete pages (Ríos-Vila, Calvo-Zaragoza, and Rizo 2023; Ríos-Vila et al. 2026). Yet the apparent progress hides a deployment problem: a model trained on one renderer, score collection, vocabulary, or page layout can fail sharply on another.

The shift is visible before any model is run. Synthetic piano pages have clean backgrounds, regular spacing, and con-

strained engraving, whereas scanned historical scores contain uneven contrast, page texture, warped systems, typography changes, and more varied annotations. The output space shifts as well: Humdrum-based encodings, Linearized MusicXML (LMX), tuplets, voices, and staff markers differ in vocabulary and structural assumptions. General transfer-learning results predict that higher model layers become increasingly task-specific (Pan and Yang 2010; Yosinski et al. 2014); for OMR, the decoder and notation vocabulary are precisely the most domain-specific components.

This paper introduces RIGOR-OMR, short for *Relation-aware Interpretable Geometry for Out-of-domain Recognition*. Rather than relying exclusively on a source-specific autoregressive vocabulary, RIGOR-OMR exposes a modular recognition path: staff geometry establishes a normalized coordinate frame; DEIM-D-FINE detects symbols; optional SAM2 masks refine local shape evidence; explicit relations connect noteheads, stems, and beams; and a deterministic backend exports MusicXML or an evaluation-compatible sequence. The intermediate artifacts make it possible to distinguish detection, pitch, duration, relation, and serialization failures.

Our contributions are fourfold:

- We formulate practical OMR applicability as zero-shot transfer under matched inputs, frozen target-domain decisions, and a common content projection, rather than extrapolating from in-domain accuracy.
- We introduce RIGOR-OMR, an inspectable geometry–detection–relation–reconstruction pipeline whose detector can be replaced without changing the structured backend.
- Across Polish and Debussy, local runs against SMT and Zeus show that RIGOR-OMR is significantly stronger than both baselines on Polish and significantly stronger than SMT while statistically comparable to Zeus on Debussy.
- Full-factorial and edge-level ablations isolate explicit relation reasoning as the main transferable mechanism, while strict OLiMPiC metrics identify global ordering and hierarchy as the remaining limitation.

2 Related Work

2.1 Datasets and Modular OMR

OMR research has historically decomposed a page into staff detection, symbol recognition, and notation reconstruction

(Calvo-Zaragoza, Jr., and Pacha 2021; Rebelo et al. 2012). Dataset design reflects this decomposition. MUSCIMA++ provides handwritten symbols and explicit pairwise relations, enabling notation-graph experiments (Jr. and Pecina 2017). DeepScores and DeepScoresV2 provide dense synthetic annotations for tiny printed objects, instance segmentation, and detection (Tuggenier et al. 2018, 2021). Camera-PrIMuS supports end-to-end recognition under realistic image degradation but is primarily monophonic (Calvo-Zaragoza and Rizo 2018). RealScores and recent instance-segmentation studies further emphasize that real-world score images differ materially from synthetic renders (Shatri and Fazekas 2023, 2024).

Object-centric OMR benefits from detectors that preserve small, dense instances. DETR introduced set prediction with bipartite matching (Carion et al. 2020); RT-DETRv2, D-FINE, and DEIM improve multiscale sampling, localization, and convergence (Lv et al. 2024; Peng et al. 2025; Huang et al. 2025). Promptable segmentation offers a complementary route from boxes to boundaries. SAM established zero-shot promptable segmentation (Kirillov et al. 2023), while SAM2 extended the design and improved image efficiency (Ravi et al. 2025). RIGOR-OMR uses these models as reusable components, not as evidence that natural-image priors alone understand notation semantics. DINOv2 features support retrieval and crop-level auxiliary classification (Oquab et al. 2024), but all final claims are based on end-output metrics rather than embedding similarity.

2.2 End-to-End Pianoform Recognition

GrandStaff introduced a large pianoform corpus and a sequence-recognition formulation over kern-like encodings (Ríos-Vila, Calvo-Zaragoza, and Rizo 2023). Sheet Music Transformer extended image-to-sequence OMR beyond monophonic transcription (Ríos-Vila, Calvo-Zaragoza, and Paquet 2024), and full-page variants combined convolutional encoders, autoregressive decoders, synthetic curriculum learning, and target-domain fine-tuning (Ríos-Vila et al. 2026). TrOMR likewise targets polyphonic recognition with Transformer-based decoding (Li et al. 2023). OLIMPiC introduced LMX, aligned MusicXML-compatible output, scanned evaluation systems, and a tree-aware metric (Mayer et al. 2024). These systems demonstrate strong in-domain accuracy, but their results also motivate a controlled view of cross-domain behavior.

2.3 Transfer and Domain Shift

Transfer learning studies distinguish reusable lower-level representations from source-specific higher layers (Pan and Yang 2010; Yosinski et al. 2014). Domain-adversarial training seeks features that cannot discriminate source from target domains (Ganin et al. 2016), and domain-generalization work organizes interventions into data, representation, and learning strategies (Wang et al. 2021). RIGOR-OMR studies a narrower and practically common setting: an OMR system must operate on an unseen collection without target annotations, while appearance, layout, and output representation may shift simultaneously.

3 Methodology

Figure 1 summarizes the pipeline. Each stage writes inspectable artifacts, allowing an end-output error to be traced to geometry, detection, relation assignment, or structured export.

3.1 Staff Geometry Estimation

Given a score image I , horizontal projections and connected components identify five-line staff groups. For staff k , the line coordinates $\{\ell_{k,j}\}_{j=1}^5$ define the robust spacing

$$s_k = \text{median}_{j=1}^4 (\ell_{k,j+1} - \ell_{k,j}). \quad (1)$$

The lines and inter-line half spaces form a discrete pitch grid G_k . We express box sizes, attachment distances, and prompt offsets in units of s_k , so the downstream rules remain comparable across resolution and engraving scale. The geometry stage outputs staff regions, line coordinates, normalized symbol coordinates, and candidate pitch-grid positions.

3.2 Staff-Aware Symbol Detection

The score image is processed by a 58-class DEIM-D-FINE detector. Its HGNetv2 backbone and hybrid encoder produce multiscale features, while symbol queries predict class-labeled boxes (Huang et al. 2025; Peng et al. 2025). Depending on page shape, inference uses either the resized full page or overlapping page-coordinate tiles; predictions are merged by class-aware non-maximum suppression. Each detection $d_i = (b_i, c_i, q_i)$ contains a box b_i , symbol class c_i , and confidence q_i ; overlap with the estimated staff regions assigns it to a staff and maps its center to normalized grid coordinates. The frozen runs use a separate four-way clef ROI classifier where enabled; it cannot relabel non-clef symbols.

3.3 Staff-Aware Prompt Generation and Mask Refinement

For each detection d_i , the frozen main run passes its unexpanded box b_i to SAM2.1 Hiera-Tiny as a box-only prompt (Ravi et al. 2025). Positive points and staff-line negative points are reserved for prompt-strategy ablations and are not part of the reported default. SAM2 is frozen and acts as an optional boundary refiner rather than a notation classifier; this distinction matches the factorial ablation in Section 4.3, where masks improve inspectability but not the final Event-F1.

3.4 Geometric Primitive Extraction and Relation Parsing

Each refined instance mask is converted into an appropriate primitive: polygons for area-like noteheads and accidentals, and pruned skeletons or endpoints for stems, beams, ledger lines, barlines, and slurs. Every primitive becomes a typed node $v \in V$ with class, staff index, box, mask evidence, and staff-normalized coordinates. RIGOR-OMR then constructs $G = (V, E)$ using class-compatible geometric candidates. Notehead–stem edges require horizontal compatibility and a bounded endpoint distance; stem–beam edges require proximity between a stem endpoint and beam support. Ledger–notehead, accidental–notehead, and slur/tie endpoint

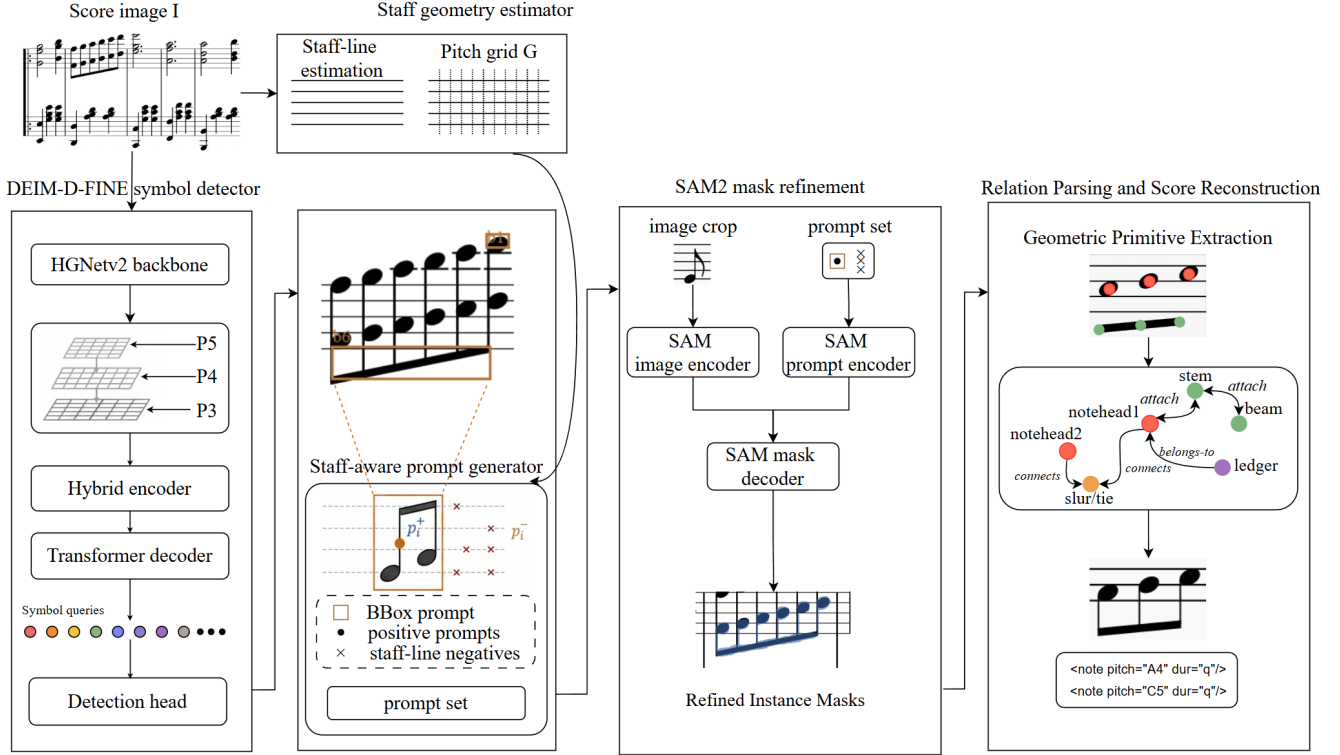


Figure 1: Overview of RIGOR-OMR. A staff-geometry estimator defines a normalized pitch grid; DEIM-D-FINE detects notation symbols; staff-aware prompts guide optional SAM2 mask refinement; and typed geometric relations support pitch–duration event assembly and structured score reconstruction.

candidates use analogous type-specific constraints. Conflict pruning retains a consistent set of attachments before event assembly.

3.5 Global Structure Decoder

The upgraded reconstruction backend converts connected primitives into musical events and then decodes a canonical score order; it is a deterministic backend upgrade rather than an additional learned model. The clef-conditioned grid G_k determines pitch, while stem, beam, flag, and dot evidence determines duration, and relation edges bind these attributes to the same event. The decoder groups staves into piano systems, obtains shared measure partitions from supported barlines, clusters staff-normalized horizontal coordinates into onset hypotheses, groups compatible simultaneous notes into chords, and serializes in system–measure–onset–staff order with deterministic pitch and geometry tie breaking. It exports both an auditable event representation and MusicXML or a dataset-compatible proxy sequence. Because decoding is deterministic once V and E are fixed, relation families can be removed and the score rebuilt without retraining or rerunning the detector, enabling the controlled edge ablations in Section 4.3.

4 Experiments

4.1 Protocol and Baselines

We study transfer from a source domain $\mathcal{D}_s$ to held-out target test sets $\mathcal{D}_t$. No target test image is used for training or checkpoint selection; dataset adapters and run-time settings are fixed before test evaluation and disclosed below. Because the methods emit heterogeneous sequences, every valid output is projected to a multiset of pitch–duration events. For multiset intersection M and predicted/reference counts P, G , we report

$$\text{EventF1} = 100 \frac{2M}{P + G}. \quad (2)$$

This order-invariant measure isolates content recovery; SER and normalized tree-edit distance (TEDn) additionally assess ordering and hierarchy on OLiMPiC. All methods receive identical images and deterministic preprocessing, failed outputs remain in the denominator, and reported baseline values come from local inference. Differences use 10,000 paired bootstrap resamples over systems/pages and are significant only when the 95% interval excludes zero.

Table 1 summarizes the target domains and evaluation roles. We compare RIGOR-OMR with the released FP-GrandStaff checkpoint of SMT (Ríos-Vila, Calvo-Zaragoza, and Paquet 2024) and the GrandStaff-LMX checkpoint of Zeus (Mayer et al. 2024).

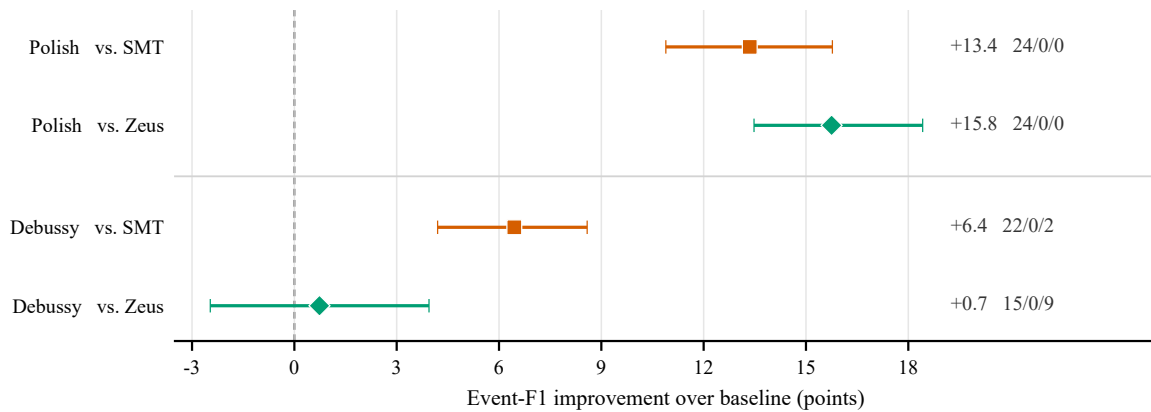


Figure 2: Paired Event-F1 improvements over each baseline. Bars are 95% bootstrap intervals; annotations give the mean difference and wins/ties/losses.

Table 1: Evaluation domains, sizes, and experimental roles.

Dataset	Target domain	N	Role
Polish	historical scans	24	comparison
Debussy	unseen editions	24	comparison/ablation
GrandStaff	clean rendered	200	relation transfer
OLiMPiC	mixed systems	200	strict/relations

4.2 Comparison

Table 2: Zero-shot pitch–duration content recovery on Polish and Debussy.

Target	Method	Prec.	Rec.	F1	Pred./Gold
Polish	RIGOR-OMR	18.734	15.700	17.083	0.838
	SMT	2.733	5.879	3.731	2.151
	Zeus	3.971	0.800	1.332	0.202
Debussy	RIGOR-OMR	7.058	10.737	8.517	1.521
	SMT	1.277	5.432	2.068	4.252
	Zeus	7.463	8.126	7.781	1.089

Table 2 gives the central comparison. On Polish, RIGOR-OMR recovers 1,589 matching events and reaches 17.083 Event-F1, improving over SMT and Zeus by 13.352 and 15.751 points with intervals $[10.891, 15.774]$ and $[13.477, 18.422]$. On Debussy it reaches 8.517, exceeding SMT by 6.449 points ($[4.200, 8.585]$) and remaining comparable to Zeus. The output ratios expose complementary failure modes: SMT over-generates on Debussy (4.252), whereas Zeus under-generates on Polish (0.202); RIGOR-OMR remains closer to the reference volume on both targets.

The aggregate intervals do not show how consistently the gain appears across individual pages or systems. Figure 3 therefore retains the paired item-level differences. Polish separates cleanly from both baselines; on Debussy, RIGOR-OMR consistently exceeds SMT, whereas the comparison with Zeus contains both wins and losses and overlaps zero at corpus level.

Order-invariant recovery does not establish globally correct notation. Table 3 therefore reports strict OLiMPiC evaluation.

RIGOR-OMR completes all 200 conversions and outperforms SMT, but Zeus remains substantially stronger in SER, Event-F1, and TEDn. Thus the transferable local geometry produces evaluable structure without solving global ordering or hierarchy.

Table 3: Strict OLiMPiC test200 results. Lower SER/TEDn and higher Event-F1 are better.

Method	SER	No tup.	Event-F1	TEDn
RIGOR-OMR	74.421	73.574	8.786	61.342
SMT	100.000	–	0.000	100.000
Zeus	56.486	50.625	48.242	47.751

4.3 Ablation

The Debussy factorial evaluates staff geometry (A), detection (B), SAM2 refinement (C), and relation reasoning (D). A and B form a non-separable perception gate: A assigns instances to staff-normalized coordinates needed for pitch and grouping, while B supplies those instances. Without either, no valid pitch–duration event can be formed. The 12 configurations with $A = 0$ or $B = 0$ are therefore structurally equivalent zero-output cases and are losslessly collapsed into one row of Table 4; all four valid $AB \times C \times D$ combinations remain visible.

Table 4: Debussy factorial ablation. A/B form the perception gate; C is SAM2 and D is relation reasoning.

Variant	Prec.	Rec.	F1	Pred./Gold
$AB+C+D$	7.058	10.737	8.517	1.521
$AB+D$	7.113	10.695	8.544	1.504
$AB+C$	8.437	2.568	3.938	0.304
AB	8.437	2.568	3.938	0.304
$A=0$ or $B=0$ (12)	0.000	0.000	0.000	0.000

With the full $AB+C+D$ path, Event-F1 is 8.517. Removing D lowers it to 3.938 (-4.579 , paired interval $[2.288, 7.019]$) as recall falls from 10.737 to 2.568, showing that isolated symbols do not recover complete events. Removing C instead changes 8.517 to 8.544 ($+0.027$,

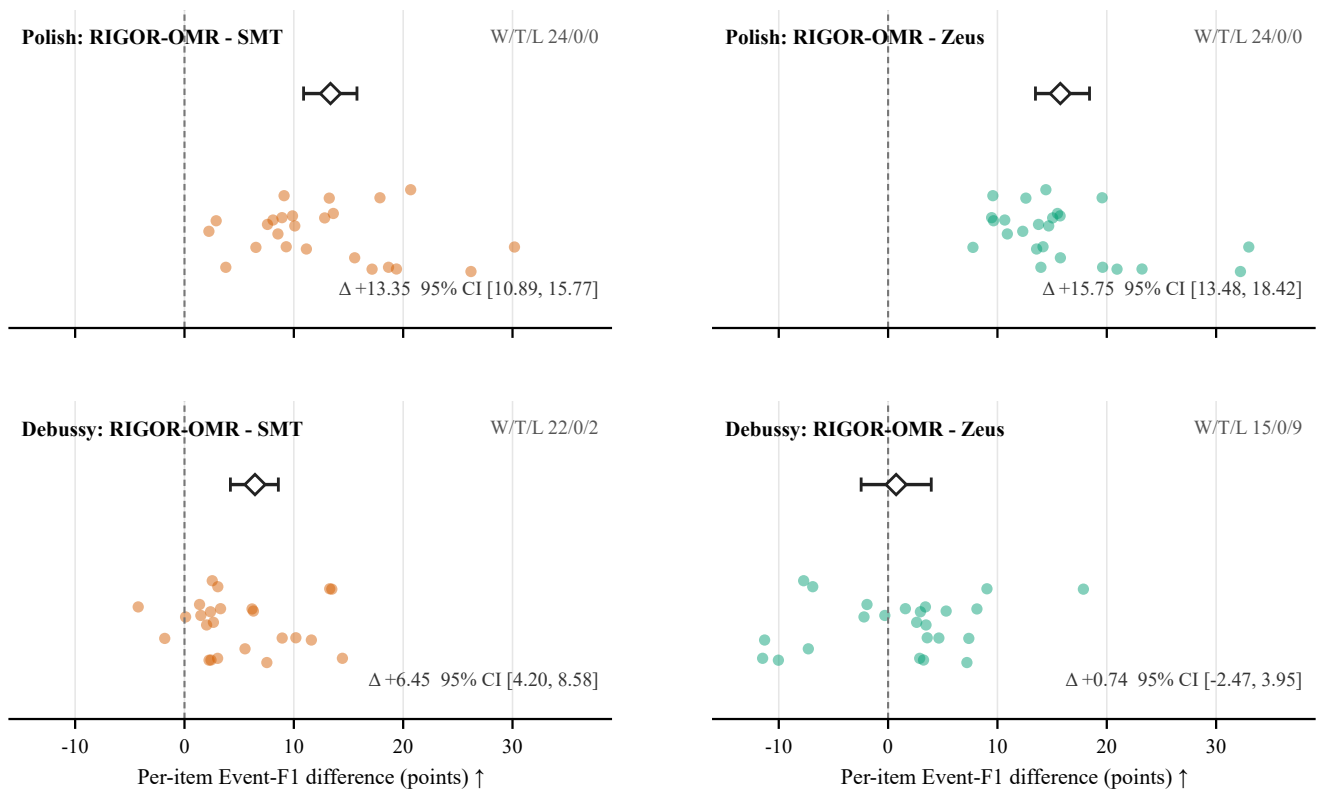


Figure 3: Per-item zero-shot Event-F1 differences. Positive values favor RIGOR-OMR; dots are held-out pages or systems, and diamonds/bars give the corpus difference and paired 95% bootstrap interval.

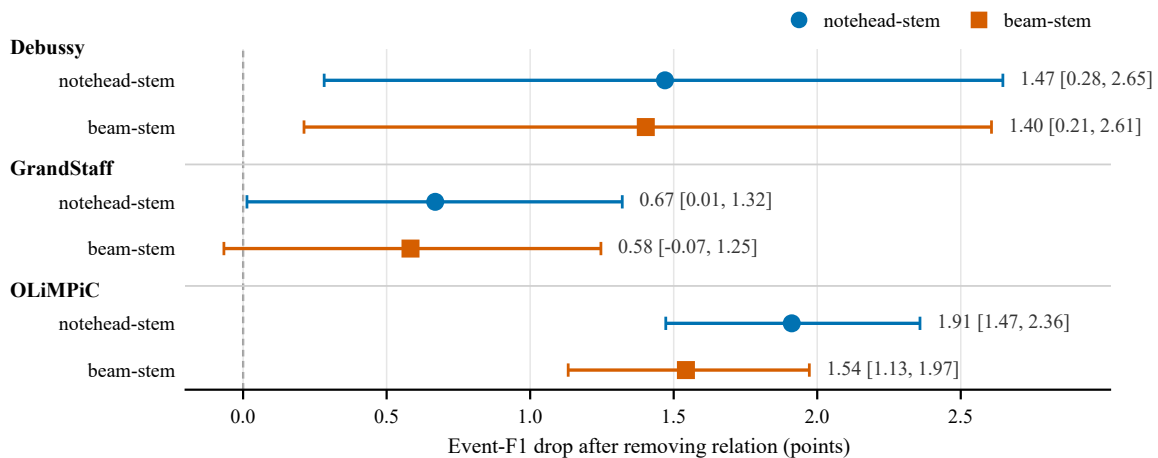


Figure 4: Event-F1 decrease after removing a relation family from frozen full-pipeline outputs. Points are corpus differences and bars are paired 95% intervals.

[−0.614, 0.793]). SAM2 therefore supplies inspectable geometry, while the verified endpoint gain comes from relation reasoning.

We next remove one relation family from cached full-pipeline outputs and deterministically reconstruct without rerunning vision. Figure 4 visualizes the effect and Table 5 retains the complete scores. Notehead–stem edges transfer

most consistently; beam–stem edges improve Debussy and OLiMPiC and have a positive but interval-overlapping effect on GrandStaff. Slur/tie removal is neutral under the pitch–duration projection.

Table 5: Complete relation ablation under frozen visual predictions. Δ is full minus ablated Event-F1; bold intervals exclude zero.

Target Removed	Full	—edge	Δ [95% CI]
Deb. note—stem	8.517	7.047	1.470 [0.282,2.647]
Deb. beam—stem	8.517	7.114	1.403 [0.212,2.607]
Deb. slur/tie	8.517	8.517	0.000 [0.000,0.000]
GS note—stem	8.887	8.217	0.669 [0.013,1.321]
GS beam—stem	8.887	8.304	0.583 [−0.067,1.246]
OLi. note—stem	9.624	7.713	1.911 [1.472,2.358]
OLi. beam—stem	9.624	8.082	1.543 [1.132,1.972]

Table 6: Frozen relation inventory as edge count / affected pages.

Target	Note—stem	Beam—stem	Slur/tie	Ledger—note
Deb.-24	3900/24	585/23	128/21	inactive
GS-200	14947/200	2395/192	198/83	4/4
OLi.-200	30113/200	333/111	266/117	8252/200

All edge ablations reuse frozen visual outputs and deterministically rebuild the prediction. Debussy intervals use 10,000 resamples with seed 20260720; both 200-item reproduction gates complete without failed pages. The ablation-specific OLiMPiC full score is not pooled with the official-style score in Table 3.

4.4 Implementation and Reproducibility

Table 7 records the settings needed to distinguish the reported runs. They share the same OMR-trained DEIM-D-FINE checkpoint, 58-class taxonomy, staff geometry, and relation code, while page inference and maximum retained detections are logged per corpus rather than presented as identical. The detector operates at 640×640 , uses score threshold 0.05, and merges overlapping predictions with class-wise NMS at IoU 0.50.

Table 7: Frozen implementation settings for the reported modular runs.

Component	Setting
Detector	DEIM-D-FINE; 58 classes; 640×640 input
Filtering	score 0.05; class-wise NMS IoU 0.50
Debussy	auto page mode; at most 1,000 detections
GrandStaff	1280-px tiles; 0.25 overlap; at most 2,000
Mask refiner	frozen SAM2.1 Hiera-Tiny; box-only prompt
Reconstruction	deterministic typed local rules

Every fused box is processed by frozen SAM2.1 Hiera-Tiny with the box-only prompt used by the main runs. Masks provide area, orientation, skeleton, and endpoint measurements; a failed or empty refinement falls back to box geometry. Relation construction is local and deterministic: notehead—stem candidates use a distance threshold $\max(8 \text{ px}, 1.35s)$, and beam—stem candidates use $\max(5 \text{ px}, 0.45s)$ around an expanded beam support. A beam group is retained only with at least two stems. No Hungarian matching, integer program, message passing, or graph neural network is used.

4.5 Qualitative and Diagnostic Analysis

Figure 5 follows the best Debussy system (test_0006, Event-F1 24.473). Its crop exposes staff localization, two noteheads, two geometry-derived stems, one beam, SAM2 masks, and the stored graph. Reconstruction composes notehead—stem and beam—stem edges, without a direct notehead—beam predictor.

Figure 6 separates recovery from binding. Pitch/duration F1 is 26.119/33.901 for RIGOR-OMR, 4.553/16.530 for SMT, and 20.318/29.550 for Zeus; count-correct scores are 11.118, 5.562, and 9.234. Yet joint F1 is 8.517: 527 events have correct pitch but unmatched duration, and 760 the converse, leaving exploitable binding headroom.

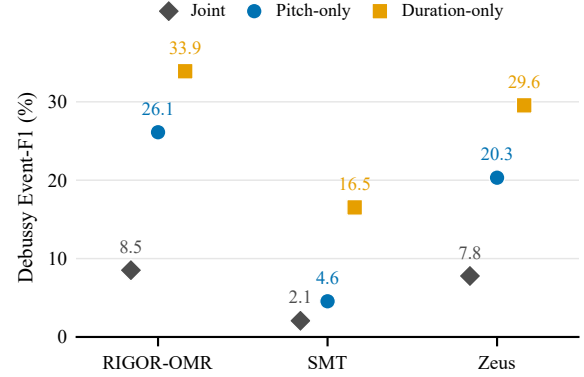


Figure 6: Joint, pitch-only, and duration-only Event-F1 on Debussy; marginal recovery reveals binding headroom.

Table 8: Retained artifacts used to audit the modular prediction path.

Stage	Retained evidence	Audited role
Staff	lines, spacing, regions	normalization
Detection	class, box, confidence	symbol proposal
Mask	binary mask, descriptors	shape/endpoints
Relation	type, endpoints, score	event binding
Decoder	semantic JSON, MusicXML	metric/export

The stored artifacts in Table 8 make Figure 5 reproducible from intermediate outputs rather than from a hand-drawn overlay. Mask postprocessing clips each prediction to its class prompt box, removes only undersized eight-connected components, and retains every surviving component intersecting that box. If filtering leaves no component, the pipeline falls back to the prompt mask and never drops a symbol solely because SAM2 failed. Each accepted relation stores its type, endpoints, and geometry score; detection confidence does not enter candidate scoring, while an assembled note confidence averages its notehead confidence and selected stem-relation score. Event-F1 is then recomputed from exported MusicXML as a multiset intersection over pitch tokens (including rest token R) and rational durations $\text{duration/divisions}$; duplicates and failed pages therefore remain auditable.

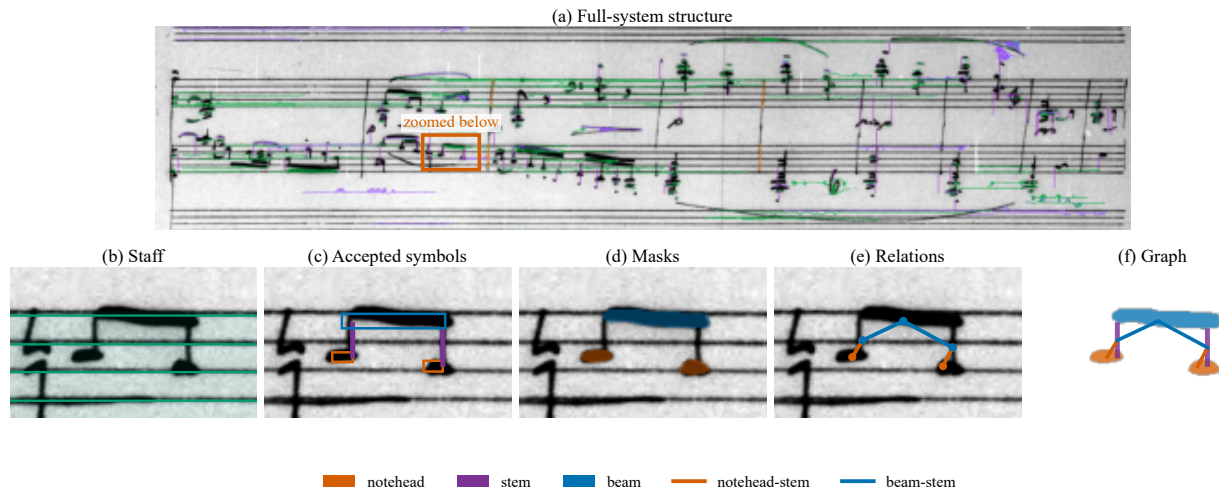


Figure 5: Debussy test_0006: (a) full-system overlay and selected crop; (b–f) staff geometry, accepted symbols, SAM2 masks, typed relations, and graph reconstruction. Masks are inspectable intermediates, not endpoint evidence.

Polish diagnostics locate a related bottleneck. Canonical ordering lowers SER by only 0.1778 points (96.1367 to 95.9589); Table 9’s privileged rows are non-deployable lower bounds. Exact-onset R3 reaches 95.9293 SER but covers only 141/5,002 gold groups. The 57.0793 cross-page bag value permits invalid cross-page matching and is only a mathematical reference.

Table 9: Polish structural diagnosis. † denotes privileged page-constrained lower bounds, not deployable outputs.

Setting	SER↓	Gain	Status
Canonical R0	95.96	0.00	observed
Staff-constrained†	91.75	4.21	diagnostic
Page-constrained†	85.26	10.70	diagnostic
Rebinding†	77.20	18.76	diagnostic

The lower bounds relax one constraint at a time under page-local target annotations. Staff-constrained matching preserves predicted staff membership; page-constrained matching allows reassignment within the same page; rebinding additionally finds the best pitch–duration pairing while preserving the predicted pitch and duration marginals. These rows identify where errors reside but cannot be executed at inference time. System-level diagnosis is unavailable because the Polish transcription does not expose page-system boundaries, and voice accuracy cannot be measured because the current prediction semantics contain no voice field. Moreover, R3 covers only 2.8% of gold onset groups, so its near-zero aggregate change cannot be interpreted as evidence that onset order is solved. Chord-cell structure is likewise normalized away by the proxy SER. We therefore use these quantities only to separate ordering, assignment, and binding headroom from the deployable canonical output.

Transferable geometry and typed relations explain the gain; global assignment and pitch–duration binding remain the main opportunities.

4.6 Limitations and Responsible Use

Absolute Event-F1 remains low and is not deployment-ready. Event-F1 parses each generated MusicXML file into an unordered multiset of pitch token and rationalized duration; it omits measure, staff, voice, reading order, chord, and slur structure, which matter on strict OLIMPIC. SAM2 has no verified endpoint gain; slur/tie edges are inactive under this projection; and 24-item Polish/Debussy tests require broader evaluation. Dataset-specific serialization and run settings also prevent pooling every score into a single protocol. The draft exporter defaults unspecified key, time, and voice, does not balance measure durations, does not decode tuplets, and does not distinguish ties from slurs; generated MusicXML is therefore an audit artifact, not an engraving-ready score. Matched-hardware efficiency comparisons are unavailable. Licensed sources, confidence disclosure, retained boxes/masks/relations, and human correction remain necessary for archival use.

5 Conclusion

RIGOR-OMR improves zero-shot recovery through staff-normalized geometry and typed relations, outperforming both baselines on Polish and SMT on Debussy. Frozen relation ablations consistently support notehead–stem reasoning and show complementary beam–stem gains, while the Debussy audit trail exposes how detections become masks, typed edges, and events. Strict OLIMPIC nevertheless shows that local correctness has not yet produced globally ordered notation; constrained binding, explicit voice modeling, and measure-consistent reconstruction are the next steps.

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